

## Chapter – 2

### Tira Online Beauty & Cosmetics E-Commerce Database

#### 2.0 Introduction

The entities required for the Tira Online Beauty & Cosmetics Shopping Database are identified based on the business requirements of the online beauty shopping system. The attributes of each entity, along with their primary keys, foreign keys, unique keys, and attribute types, are determined to ensure proper data organization. The relationships between the entities are also identified to represent the interactions within the system and to establish a well-structured relational database design. In addition, the cardinality and participation constraints of each relationship are analyzed to define how entities are associated with one another. This analysis provides a clear understanding of the data requirements and serves as the foundation for designing the Entity Relationship (ER) diagram, relational schema, and subsequent database implementation while ensuring data integrity, consistency, and minimal redundancy.

#### 2.1 Entity: Customer

| Attribute     | Key Type    |
|---------------|-------------|
| Customer_ID   | Primary Key |
| First_Name    | Not Null    |
| Last_Name     | -           |
| Email         | Unique Key  |
| Phone         | Unique      |
| Date_of_Birth | -           |
| Gender        | -           |
| Address       | Not Null    |

#### Entity: Brand

| Attribute   | Key Type    |
|-------------|-------------|
| Brand_ID    | Primary Key |
| Brand_Name  | Unique Key  |
| Country     | -           |
| Description | -           |

#### Entity: Category

| Attribute     | Key Type    |
|---------------|-------------|
| Category_ID   | Primary Key |
| Category_Name | Unique Key  |

|             |   |
|-------------|---|
| Description | - |
|-------------|---|

### Entity: Product

| Attribute    | Key Type    |
|--------------|-------------|
| Product_ID   | Primary Key |
| Product_Name | Not Null    |
| Brand_ID     | Foreign Key |
| Category_ID  | Foreign Key |
| Price        | Not Null    |
| Shade        | -           |
| Skin_Type    | -           |
| Expiry_Date  | -           |
| Stock        | -           |

### Entity: Cart

| Attribute   | Key Type    |
|-------------|-------------|
| Cart_ID     | Primary Key |
| Customer_ID | Foreign Key |
| Total_Items | -           |
| Grand_Total | -           |

### Entity: Order

| Attribute    | Key Type    |
|--------------|-------------|
| Order_ID     | Primary Key |
| Customer_ID  | Foreign Key |
| Order_Date   | -           |
| Order_Status | -           |
| Total_Amount | -           |

### Entity: Order\_Item

| Attribute    | Key Type    |
|--------------|-------------|
| OrderItem_ID | Primary Key |
| Order_ID     | Foreign Key |
| Product_ID   | Foreign Key |
| Quantity     | -           |
| Unit_Price   | -           |

### Entity: Payment

| Attribute      | Key Type    |
|----------------|-------------|
| Payment_ID     | Primary Key |
| Order_ID       | Foreign Key |
| Payment_Method | -           |

|                |   |
|----------------|---|
| Payment_Status | - |
| Payment_Date   | - |
| Amount         | - |

### Entity: Review

| Attribute   | Key Type    |
|-------------|-------------|
| Review_ID   | Primary Key |
| Customer_ID | Foreign Key |
| Product_ID  | Foreign Key |
| Rating      | -           |
| Review_Text | -           |
| Review_Date | -           |

### Entity: Delivery

| Attribute        | Key Type    |
|------------------|-------------|
| Delivery_ID      | Primary Key |
| Order_ID         | Foreign Key |
| Delivery_Address | -           |
| Delivery_Date    | -           |
| Delivery_Status  | -           |

## 2.2 Entity Relationships

| Entity 1 | Relation     | Entity 2   | Cardinality  | Participation     |
|----------|--------------|------------|--------------|-------------------|
| Customer | Owns         | Cart       | One-to-One   | Partial / Total   |
| Customer | Places       | Order      | One-to-Many  | Partial / Total   |
| Customer | Writes       | Review     | One-to-Many  | Partial / Total   |
| Brand    | Manufactures | Product    | One-to-Many  | Partial / Total   |
| Category | Contains     | Product    | One-to-Many  | Partial / Total   |
| Cart     | Contains     | Product    | Many-to-Many | Partial / Partial |
| Order    | Includes     | Order_Item | One-to-Many  | Total / Total     |
| Product  | Appears In   | Order_Item | One-to-Many  | Partial / Total   |
| Order    | Has          | Payment    | One-to-One   | Total / Total     |
| Order    | Has          | Delivery   | One-to-One   | Total / Total     |
| Product  | Receives     | Review     | One-to-Many  | Partial / Total   |